Arijit (Percy) Chakma

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Education

Drexel University, Philadelphia, PA — Expected June 2027
B.S. in Computer Science, Minor in Mathematics / *GPA:* ***3.80/4.00***
Coursework: Deep Learning (graduate), Applications of Machine Learning (graduate), Machine Learning, Computational Photography, Augmented Reality Programming, Optimization Theory (Linear Programming).

Research Interests

My research lies at the intersection of generative AI, multimodal learning, and human–AI interaction. I am interested in building controllable and reliable AI systems that reason about visual and spatial information, and in how they can support intelligent interfaces, embodied agents, and real-world applications.

Publications

[1] **A. Chakma**, P. He, Z. Wang, H. Liu, T. Li, T. D. Do, F. Liu. "DrawSim-PD: Simulating Student Science Drawings to Support NGSS-Aligned Teacher Diagnostic Reasoning." *International Conference on Artificial Intelligence in Education (AIED)*, 2026. [DOI] [Project]

[2] O. A. Khan, N. Gomez, **A. Chakma**, T. D. Do. "Augmented Reality Hand Outlines Modulate Weight Perception of Real Objects." *Under review at ACM CHI*, 2027.

[3] **A. Chakma**, F. Liu. "Encoded Where It Is Drawn, Absent Where It Is Read: Relational Evidence in Scientific Diagrams." *Manuscript*, 2026. [PDF]

Research Experience

Undergraduate Researcher, Visual Intelligence Lab, Drexel University — Apr. 2025 – present
Advisor: Prof. Feng Liu

- Led **DrawSim-PD** [1], a framework for simulating K–12 science drawings with **controllable misconceptions**; designed its multi-stage pipeline combining LLMs, VLMs, diffusion models, and structured concept representations to produce NGSS-aligned drawings across levels of student understanding.
- Generated **10,000** student-like drawings spanning **100** science topics.
- Ran expert educator evaluations of the drawings' conceptual alignment, realism, and diagnostic usefulness.
- Built controlled graph-based stimuli to study how **diffusion transformers** (Qwen-Image, SD3.5) encode relations in scientific diagrams [3].
- Designed **representation probes** to locate where these models encode relational information.
- Ran **causal interventions** with attention-knockout controls to test how they affect generated diagram structure.
- Developing a **failure-aware synthetic data framework** for monocular 3D object detection on KITTI that diagnoses detector failures and generates targeted training examples for them.

Undergraduate Researcher, DIVA Lab, Drexel University — Aug. 2026 – present
Advisor: Prof. Tiffany D. Do / *Mentor: Omar A. Khan*

- Developed a real-time grasp-detection pipeline integrating YOLOv8s and MediaPipe with Meta Quest 3 passthrough for an AR study of how virtual hand outlines change the perceived weight of real objects [2].
- Developing a gaze-aware, voice-enabled AI coding assistant for VS Code that uses eye tracking, code edits, and execution feedback to detect developer struggle and give spoken guidance via an LLM agent.

Honors and Awards

- **Drexel Global Scholarship** – full ride (**$330,000**), selection rate under **0.5%** — 2022–2027
- **1st Place Overall** (of 105 projects), PennApps, University of Pennsylvania — 2024
- **2nd Place Overall** (of 80 projects), HopHacks, Johns Hopkins University — 2024
- **Best Social Hack** (of 33 projects), DragonHacks, Drexel University — 2025

- **Best Beginner Quantum Hack**, Bitcamp, University of Maryland 2025
- **Best Beginner Hack** (of 34 projects), DivHacks, Columbia University 2025
- **3rd Place, Best Philanthropic Hack** (of 19 projects), HopHacks, Johns Hopkins University 2024
- **Track Winner, Interactive & Multimedia**, DragonHacks, Drexel University 2024
- **Dean's List**, Drexel University 2022–2026

Teaching Experience

Undergraduate Course Assistant, Drexel University Sep. 2025 – present

- CS 281: Systems Architecture and CS 164: Introduction to Programming (Fall 2026); CS 383: Machine Learning (Winter 2026); CS 380: Introduction to AI (Fall 2025).
- Supported **400+ students** across four course offerings through weekly office hours, code debugging, and one-on-one concept explanations.

Industry Experience

Software Engineer Co-op, Kroll Bond Rating Agency, New York, NY Apr. – Sep. 2026

- Shipped **30+** full-stack tickets using Next.js, TypeScript, C#, and ASP.NET on a private-credit deals platform, owning UX and accessibility for analyst-facing screens.
- Prototyped a graph explorer of lender, borrower, and sponsor ties across **15,000+** deals; combined five sequential queries into one, reducing worst-case load time by **7**× (1.07 s → 0.15 s).
- Built a Campaign Monitor data pipeline enabling A/B testing across **40+** campaigns.

Cloud Software Engineer Co-op, Bristol Myers Squibb, Princeton, NJ Apr. – Sep. 2024

- Built **5+** ETL pipelines using AWS Lambda and Step Functions over **10,000+** employee records, and migrated **1.3 TB** from DynamoDB to Athena.
- Automated IAM offboarding for **1,250+** users, reducing manual effort by **90%**.

Selected Projects

TickerMaster – *Multi-Agent Trading Sandbox* [GitHub] [Devpost]

- Built a sandbox of **LLM trading agents** whose personas react to market volatility, breaking news, and each other, alongside cited per-ticker research briefs, so users can test strategies before trading real money.

WoundWatch – *AI-Powered 3D Wound Analysis* [GitHub] [Devpost]

- Built a React Native app and REST backend that **reconstructs 3D wound models from 2D photos** using Depth Anything V2 and SAM for remote clinical triage.

SoundShield – *Wearable Audio & Vision Node* [GitHub] [Devpost]

- Built a wearable Raspberry Pi node combining audio processing and computer vision to **adaptively suppress ambient noise** for people with auditory hypersensitivity.

DenEyes – *Robotic Eye-Tracking System* [GitHub] [Devpost]

- Built a vision-based robotic system that tracks eye motion and attention patterns as a **screening signal for early ADHD detection** in children.

Skills

Programming: Python, C#, TypeScript, JavaScript, C/C++, SQL, Go, Java

ML / Vision: PyTorch, Hugging Face, OpenCV, LangChain/LangGraph, NumPy, SciPy, Weights & Biases

Software / Tools: Next.js, React, React Native, ASP.NET, AWS, Docker, Git, Unity, Linux

Spoken: English, Chakma (native); Bengali, Hindi (conversational); Japanese (intermediate); Mandarin, Korean (basic)